Assessed Problem Sheet 3 — Week 8

This problem sheet is assessed. Submit your answers via Moodle by **5pm on Thursday 5th**.

Marks per section are shown in square brackets. Be clear about what you are doing, and show your working. Use Dirac notation (or a corresponding column vector / matrix notation) for all *these questions.*

This problem sheet is out of 28 marks. Your mark will be rescaled to a mark out of 100 when uploaded to Moodle.

In this problem sheet, we are going to look at the theory behind something called "Quantum sensing with NV colour centres in diamond". This is a type of quantum technology, and in fact, I run a teaching lab on it for my CDT students. There is no new maths or concepts here, instead I am adding some extra context to help you visualise why all this "spin stuff" is so important for the world of quantum technologies.

A diamond is a crystal made up of carbon atoms. Whereas, "colour centres" are atomic-like impurities (defects) that give diamonds different colours. Here, NV spin defect consists of a Nitrogen atom occupying one site of the lattice, next to an empty site (a vacancy, hence the V in the NV). In the case of NV Centres, the diamond will have a pink hue! But that isn't all...

The NV centres have a set of unique physical characteristics which means a single spin can be initialised in a chosen quantum state; they retain quantum coherence over a relatively long time even at room temperature; and unusually, the electron spin quantum state can be read off optically, so single spins can be detected. There have already been some ground-breaking results from the LCN in terms of healthcare applications. You can read more about it here.

In terms of the technical, it is a $S = 1$ (spin one) electronic spin system, with a Hamiltonian given by:

$$\hat{H} = D_0 \hat{S}_z^2 - \gamma \hat{\mathbf{S}}.\mathbf{B}$$

The first term is a crystal field along the axis from the N to the V site. It is usually strong relative to the magnetic field B, ie $D_0 \gg \gamma B$ so the crystal field sets the z axis - z axis choice in action! Recently, there has been lots of investigation to look at what happens to the diamond if it undergoes strain. The Hamiltonian will change, a new term called the strain term must be added; in its simplest form this becomes:

$$\hat{H} = D\hat{S}_z^2 - \gamma \mathbf{S}.\mathbf{B} + E\left(\hat{S}_x^2 - \hat{S}_y^2\right)$$

Now, shall we have a look at using our knowledge in this context. . .

1. Consider an NV spin system, for which $S = 1$, in a magnetic field of magnitude B along the NV axis (i.e. along $\hat{z}$). The Hamiltonian experiences a perturbation due to strain of the form:

$$\hat{H}_S = \epsilon_y\left(\hat{S}_x\hat{S}_y + \hat{S}_y\hat{S}_x\right) + \epsilon_x\left(\hat{S}_x^2 - \hat{S}_y^2\right)$$

   (a) Express $\hat{H}_S$ in terms of raising and lowering operators for spin. [2]

   (b) Obtain the matrix of $\left(\hat{S}_x^2 - \hat{S}_y^2\right)$. [2]

   (c) Defining $r = \sqrt{\epsilon_x^2 + \epsilon_y^2}$ and $\tan\phi = \epsilon_y/\epsilon_x$ show that: [4]

$$\hat{H}_S = \frac{r}{2}\left(\left(\hat{S}_-\right)^2 e^{i\phi} + \left(\hat{S}_+\right)^2 e^{-i\phi}\right)$$

   (d) Hence obtain the eigenvalues and eigenvectors of $\hat{H}_S$ for the case $\phi = 0$. *Hint! It may be useful to re-order the basis so it clearly decouples into a 2×2 matrix and an uncoupled state.* [4]

2. Consider the situation where the $S_1 = 1$ electron spin of an NV is coupled to another NV spin, for which $S_2 = 1$. We have seen that we can add two completely different quantum angular momenta to construct a total angular momentum state $|JM\rangle$ by superposing products of the individual angular momentum states. The coefficients are called Clebsch Gordan coefficients. For our system, $|JM\rangle = \sum_{m_1}\langle S_1 m_1 S_2 m_2 \mid JM\rangle\, |S_1 m_1\rangle\, |S_2 m_2\rangle$. By using the table at the end of the sheet, complete these exercises:

   (a) Write the state $|J = 1 M = 0\rangle$ in terms of the single electron states. [2]

(b) Give the state $|S_1 m_1 = 0\rangle\,|S_2 m_2 = 0\rangle$ in the total angular momentum basis. [2]

(c) For the state $|J = 1 M = 1\rangle$, work out $\left\langle \hat{S}_{z1} \right\rangle$. [2]

(d) For the state $|J = 1 M = 0\rangle$, work out $\left\langle \hat{S}_{x2} \right\rangle$. [2]

3. **In this question we are going to think about a general system.** We have a system where we will look at both the Orbital Angular Momentum and Spin, together. We will not assign quantum number values for this questions, so l and s can be left unvalued.

(a) The spin angular momentum and orbital angular momentum of a particle are represented by vector operators $\hat{\mathbf{L}}$ and $\hat{\mathbf{S}}$, the total angular momentum vector being $\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$. Show that [2]

$$\hat{J}^2 = \hat{L}^2 + \hat{S}^2 + 2\hat{\mathbf{L}} \cdot \hat{\mathbf{S}} \ .$$

Re-express this in terms of raising and lowering operators $\hat{L}_\pm$ and $\hat{S}_\pm$, and $\hat{L}_z$ and $\hat{S}_z$.

(b) If the orbital and spin angular momentum quantum numbers are ℓ and s, then $\hat{L}_z$ and $\hat{S}_z$ have the eigenvalues $m\hbar$ and $m_s\hbar$, respectively. We can describe the eigenstates as $|\ell\, m, s\, m_s\rangle = |\ell, m\rangle|s, m_s\rangle$. Demonstrate that the state $|\ell\,\ell, s\,s\rangle$, in which $m = \ell$ and $m_s = s$, is an eigenstates of $\hat{J}^2$, and state what the eigenvalue is. [6]

CLEBSCH-GORDAN COEFFICIENTS

$j_1 = \frac{1}{2}, j_2 = \frac{1}{2}$ [edit]

$m = 1$

m_1, m_2 \ j	1
$\frac{1}{2}, \frac{1}{2}$	1

$m = -1$

m_1, m_2 \ j	1
$-\frac{1}{2}, -\frac{1}{2}$	1

$m = 0$

m_1, m_2 \ j	1	0
$\frac{1}{2}, -\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
$-\frac{1}{2}, \frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

$j_1 = 1, j_2 = \frac{1}{2}$ [edit]

$m = \frac{3}{2}$

m_1, m_2 \ j	$\frac{3}{2}$
$1, \frac{1}{2}$	1

$m = \frac{1}{2}$

m_1, m_2 \ j	$\frac{3}{2}$	$\frac{1}{2}$
$1, -\frac{1}{2}$	$\sqrt{\frac{1}{3}}$	$\sqrt{\frac{2}{3}}$
$0, \frac{1}{2}$	$\sqrt{\frac{2}{3}}$	$-\sqrt{\frac{1}{3}}$

$j_1 = \frac{3}{2}, j_2 = \frac{1}{2}$ [edit]

$m = 2$

m_1, m_2 \ j	2
$\frac{3}{2}, \frac{1}{2}$	1

$m = 1$

m_1, m_2 \ j	2	1
$\frac{3}{2}, -\frac{1}{2}$	$\frac{1}{2}$	$\sqrt{\frac{3}{4}}$
$\frac{1}{2}, \frac{1}{2}$	$\sqrt{\frac{3}{4}}$	$-\frac{1}{2}$

$m = 0$

m_1, m_2 \ j	2	1
$\frac{1}{2}, -\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
$-\frac{1}{2}, \frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

$j_1 = 1, j_2 = 1$ [edit]

$m = 2$

m_1, m_2 \ j	2
1, 1	1

$m = 1$

m_1, m_2 \ j	2	1
1, 0	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
0, 1	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

$m = 0$

m_1, m_2 \ j	2	1	0
1, −1	$\sqrt{\frac{1}{6}}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{3}}$
0, 0	$\sqrt{\frac{2}{3}}$	0	$-\sqrt{\frac{1}{3}}$
−1, 1	$\sqrt{\frac{1}{6}}$	$-\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{3}}$

For brevity, solutions with $M < 0$ and $j_1 < j_2$ are omitted. They may be calculated using the simple relations

$$\langle j_1, j_2; m_1, m_2 \mid j_1, j_2; J, M\rangle = (-1)^{J-j_1-j_2}\langle j_1, j_2; -m_1, -m_2 \mid j_1, j_2; J, -M\rangle.$$

and

$$\langle j_1, j_2; m_1, m_2 \mid j_1, j_2; J, M\rangle = (-1)^{J-j_1-j_2}\langle j_2, j_1; m_2, m_1 \mid j_2, j_1; J, M\rangle.$$